

Task 9: Network Vulnerability Scanning

Chapter 1: Introduction to Network Vulnerability Scanning

Network Vulnerability Scanning is a fundamental process in cybersecurity used to identify weaknesses, open ports, exposed services, and misconfigurations within a network. It helps security professionals understand how systems can be discovered and potentially exploited by attackers.

By scanning a network, organizations can detect unnecessary open ports, outdated services, and insecure configurations before they are misused. This proactive approach reduces the attack surface and strengthens the overall security posture of a system.

Network scanning is commonly used during penetration testing, security audits, and continuous monitoring to ensure that systems remain secure over time.

Chapter 2: Overview of Nmap

Nmap (Network Mapper) is an open-source network scanning and reconnaissance tool widely used by cybersecurity professionals. It is designed to discover hosts on a network, identify open ports, detect running services, and perform operating system detection.

Nmap is efficient, flexible, and supports a wide range of scanning techniques, making it one of the most popular tools for network enumeration. It is commonly used during the reconnaissance phase of ethical hacking and penetration testing.

Key features of Nmap include:

- Host discovery
- Port scanning
- Service and version detection
- Operating system identification
- Network security assessment

Chapter 3: Practical Setup and Scan Methodology

In this task, the local system (127.0.0.1) was scanned using Nmap to analyze exposed network services. Required services such as Apache (HTTP) and MariaDB (MySQL) were running to simulate a realistic environment.

The scanning process was carried out in the following steps:

Basic Port Scan

A basic scan was performed to identify open TCP ports on the system. This helps in identifying services that are accessible over the network.

Service Version Detection

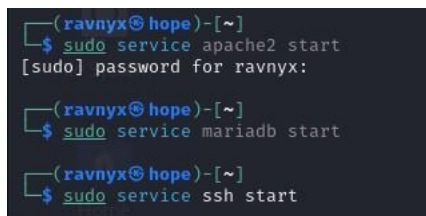
Service detection was performed to determine which applications and versions were running on the open ports. This information is crucial because outdated services often contain known vulnerabilities.

Operating System Detection

Operating system detection was carried out to identify the underlying OS running on the host. Knowing the OS helps attackers and defenders tailor their attack or defense strategies accordingly.

Figure 3.1: Starting Required Network Services

Apache, MariaDB and SSH services were started to simulate active network services before performing network vulnerability scanning.



```
(ravnyx@hope)-[~]
$ sudo service apache2 start
[sudo] password for ravnyx:

(ravnyx@hope)-[~]
$ sudo service mariadb start

(ravnyx@hope)-[~]
$ sudo service ssh start
```

Chapter 4: Analysis of Scan Results

The Nmap scan revealed multiple open ports and active services on the system:

- **Port 22 (SSH)** – OpenSSH service running, used for remote secure access.
- **Port 80 (HTTP)** – Apache web server running, hosting web applications.
- **Port 3306 (MySQL)** – MariaDB database service running.

The operating system was identified as a Linux-based system. Service version detection revealed specific software versions, which could be vulnerable if not regularly updated.

These findings indicate that the system exposes multiple services that could be targeted if not properly secured.

Figure 4.1: Basic Port Scan Using Nmap (A basic Nmap scan was performed to identify open TCP ports on the local system.)

```
(ravnyx@hope)-[~]
$ nmap 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 13:15 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds

(ravnyx@hope)-[~]
$
```

Figure 4.2: Service and Version Detection (Service detection was carried out to identify running services and their software versions.)

```
(ravnyx@hope)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 13:16 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.9p2 Debian 2 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.62 ((Debian))
3306/tcp  open  mysql    MariaDB 11.4.2
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.60 seconds

(ravnyx@hope)-[~]
$
```

Figure 4.3: Operating System Detection Using Nmap (Nmap OS detection was performed to infer the underlying operating system using TCP/IP fingerprinting techniques.)

```

(ravnyx@hope) ~
$ sudo nmap -O 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 13:19 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000052s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
3306/tcp  open  mysql
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=1/29%OT=22%CT=1%CU=32075%PV=N%DS=0%DC=L%G=Y%TM=697B110
OS:5%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=A)SEQ
OS:(SP=102%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=103%GCD=1%ISR=10A%TI=Z%
OS:CI=Z%II=I%TS=A)SEQ(SP=105%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=106%G
OS:CD=1%ISR=107%TI=Z%CI=Z%II=I%TS=A)OPS(O1=MFFD7ST11NW7%O2=MFFD7ST11NW7%O3=
OS:MFFD7NNT11NW7%O4=MFFD7ST11NW7%O5=MFFD7ST11NW7%O6=MFFD7ST11)WIN(W1=8200%W
OS:2=8200%W3=8200%W4=8200%W5=8200%W6=8200)ECN(R=Y%DF=Y%T=40%W=8200%O=MFFD7N
OS:NSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=
OS:Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+F=A
OS:R%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=4
OS:0%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=
OS:G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.87 seconds

```

Chapter 5: Risk Interpretation

Open ports and exposed services increase the system's attack surface. If services such as SSH, HTTP, or MySQL are misconfigured or outdated, attackers could exploit known vulnerabilities to gain unauthorized access.

For example:

- Weak SSH credentials can lead to brute-force attacks.
- Vulnerable web servers may allow web-based attacks.
- Exposed databases may lead to data leakage.

Regular scanning helps in early identification of such risks.

Figure 5.1: Consolidated Network Scan Results (The consolidated scan results highlight exposed ports and services, which increase the system's attack surface.)

```
(ravnyx@hope)~$ nmap -A 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 13:21 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000056s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.9p2 Debian 2 (protocol 2.0)
| ssh-hostkey:
|_ 256 f1:b4:a1:8e:b5:7c:5b:ae:61:d3:12:82:b1:08:80:d3 (ECDSA)
|_ 256 62:8e:cd:76:c6:d3:b3:78:a0:c0:e8:d6:f3:49:d5:f0 (ED25519)
80/tcp    open  http     Apache httpd 2.4.62 ((Debian))
|_ http-title: Apache2 Debian Default Page: It works
|_ http-server-header: Apache/2.4.62 (Debian)
3306/tcp  open  mysql    MariaDB 11.4.2
|_ ssl-date: TLS randomness does not represent time
|_ ssl-cert: Subject: commonName=MariaDB Server
|_ Not valid before: 2026-01-29T07:41:43
|_ Not valid after: 2036-01-27T07:41:43
|_ mysql-info:
|_ Protocol: 10
|_ Version: 11.4.2-MariaDB-4
|_ Thread ID: 34
|_ Capabilities flags: 65534
|_ Some Capabilities: ODBCClient, LongColumnFlag, Support41Auth, Speaks41ProtocolOld, SupportsCompression, IgnoreSi
gpipes, ConnectWithDatabase, IgnoreSpaceBeforeParenthesis, SwitchToSSLAfterHandshake, InteractiveClient, SupportsLoa
dDataLocal, Speaks41ProtocolNew, DontAllowDatabaseTableColumn, FoundRows, SupportsTransactions, SupportsAuthPlugins,
SupportsMultipleStatements, SupportsMultipleResults
|_ Status: Autocommit
|_ Salt: tmFRMiHlg?EzY0#],c.X
|_ Auth Plugin Name: mysql_native_password
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=1/29%OT=22%CT=1%CU=44373%PV=N%DS=0%DC=L%G=Y%TM=697B119
OS:9%P=x86_64-pc-linux-gnu)SEQ(SP=102%GCD=1%ISR=110%TI=Z%CI=Z%II=I%TS=A)SEQ
OS:(SP=104%GCD=1%ISR=108%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=107%GCD=1%ISR=109%TI=Z%
OS:CI=Z%II=I%TS=A)SEQ(SP=108%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=A)OPS(O1=MFFD7
OS:ST11NW7%O2=MFFD7ST11NW7%O3=MFFD7NNT11NW7%O4=MFFD7ST11NW7%O5=MFFD7ST11NW7
OS:%O6=MFFD7ST11)WIN(W1=8200%W2=8200%W3=8200%W4=8200%W5=8200%W6=8200)ECN(R=
OS:Y%DF=Y%T=40%W=8200%O=MFFD7NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+F=AS%
OS:RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y
OS:%DF=Y%T=40%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R
OS:%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=
OS:40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S
OS: )

Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

```
(ravnyx@hope)~$ nmap -p- 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-29 13:23 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

(ravnyx@hope)~$
```

Chapter 6: Mitigation and Security Recommendations

To reduce the risks identified during the scan, the following security measures are recommended:

- Close unused ports and services.
- Restrict access to critical services using firewalls.
- Keep all services and operating systems updated.
- Use strong authentication mechanisms for SSH and databases.
- Perform regular vulnerability scans to monitor changes in the network.

Implementing these practices significantly improves network security and resilience.

Chapter 7: Conclusion

This task demonstrated how network vulnerability scanning is performed using Nmap. By identifying open ports, services, and operating system details, potential security risks were analyzed and documented.

The exercise enhanced practical understanding of network reconnaissance and emphasized the importance of proactive security assessment in protecting systems from cyber threats.